

IS 1161:2014

CUPLOCK SYSTEM

4-LEG TOWER

SELF-DRAINING GRATING

ISO 9001:2015

The **Economy Staircase Tower** is Prime Steeltech's value-engineered cuplock staircase tower, built for projects where fast turnaround and cost control matter more than a fully sheeted finish. In place of solid anti-skid decking, the tower uses an open square-pipe grating on both the stair flights and the landing platforms, so water, mud, and loose debris fall straight through instead of collecting underfoot.

A four-leg cuplock frame carries a zig-zag stair arrangement across a 2,000 × 1,500 mm footprint, with each lift climbing 1,500 mm over six steps. The open-mesh treads make this configuration a practical fit for basement excavations, wet infrastructure sites, and any short-duration access requirement where assembly and dismantling can be done easily.

All load-bearing tube is 48.3 mm OD medium-class M.S. to IS 1161 (YST210). Vertical continuity runs through a welded 60 mm OD socket rather than a loose spigot, and the tower sits on adjustable jack bases with a 450 mm travel range so it can be levelled on broken or sloped ground without packing. Being an Economy-tier product, the Economy staircase tower ships without side rails, mid-rails, or toe boards — site teams need to plan independent fall-protection before anyone climbs it (see Section 3).

SYSTEM

Cuplock, 4-Leg

PIPE CLASS / OD

Medium · 48.3 mm

MATERIAL

M.S. (IS 1161, YST210)

TREAD / LANDING

M.S. Square & Rectangular Pipe Grating
IS(4923)

FINISH

Red-Oxide Primer + Blue Enamel

GRID SIZE

2000 x 1500



Economy deployed at a basement excavation edge



Tower rising along a building elevation



Close-up of the M.S. square-pipe grating tread

Overall Tower Geometry

GRID SIZE (L × W) 2,000 × 1,500 mm
PLATFORM HEIGHT (MAX.) Multiple of 1.5 m or 3 m
VERTICAL MEMBERS 4-leg cuplock
CUP / BAY SPACING 500 mm centers
TOP RAILING Uppermost platform only — top rail 1,500 mm, guard rail 750 mm

Stair Geometry

Stair Width (clear)	600 mm
Stair Type	Zig-zag step arm, open construction
Stair Material	M.S. square pipe grating, 25 × 25 mm
Step Riser	295 mm
Tread (going)	270 mm
Staircase Angle	54°
No. of Steps per Flight	5
Side Rail / Mid Rail	Not fitted — Economy configuration; see Section 3
Step Arm Design Load (UDL)	250 kg/m ²

Landing Platform

PLATFORM WIDTH x DEPTH 1250 mm x 450 mm
SURFACE M.S. rectangular pipe grating, 25 mm
FIXING Hook-on, secured to tower frame
TOE GUARDS Not fitted — Economy configuration
PLATFORM DESIGN LOAD (UDL) 250 kg/m ²

Structural Framing

Component	Specification
Cuplock Vertical	48.3 mm OD M.S. tube, IS 1161 Grade YST210
Top Cup	SG / malleable cast iron
Bottom Cup	Pressed metal
Cup Spacing	500 mm centers
Cuplock Ledger	48.3 mm OD M.S. pipe, forged blade each end
Vertical Joint	60 mm OD welded socket (no loose spigot pin)
Bracing	48.3 mm OD M.S. pipe, all four faces
Couplers & Clamps	Sheet-metal fabricated

Base & Anchorage

Component	Specification
Adjustable Base Jack	38 mm OD spindle · 600 mm overall · 450 mm adjustable travel · 150 × 150 × 6 mm M.S. base plate
Tie Bar structure or wall	48 mm OD pipe with welded holding plate (150 x 150 x 6 mm); anchors to RCC

All dimensions in mm unless noted. Weights and quantities subject to ±3% tolerance. Mandatory tie spacing of 3.0 m (vertical and horizontal) applies once platform height exceeds 6.0 m.

Key Features & Applications

What Sets the Economy Apart

Tool-Free Cuplock Erection

SG / malleable cast iron top cup locks up to four ledgers at a single node without spanners.

Welded Socket Joint

60 mm OD welded socket gives a more positive vertical connection than a loose spigot pin.

Tie-Bar Anchorage

48 mm OD tie bar with welded plate; mandatory above 6.0 m platform height.

Left Out of This Configuration

No Side or Mid Rails

Stair flights carry no handrail or mid-rail; a supplementary system is required for personnel safety.

Open-Mesh Grip Only

Foot grip relies on the grating pattern — there is no dimpled anti-skid sheet surface.

Where the Economy Staircase Tower Fits

- Basement excavation and below-grade access in wet or muddy ground
- Short-duration site access scaffolding with frequent relocation
- Light industrial inspection walkways and maintenance access
- Budget-led projects that supply their own independent fall-arrest system

Safety Guidelines

SAFETY NOTICE — REDUCED FALL PROTECTION. This configuration has no side rails, mid-rails, or toe boards. It should not be occupied without a supplementary system — independent guardrails, a harness and lanyard, or a continuous lifeline — and a site-specific risk assessment signed off by a competent person.

1. Erection, alteration, and inspection must be carried out by a competent person following a documented method statement.
2. Above 6.0 m platform height, tie bars are mandatory at a maximum of 3.0 m spacing in both the vertical and horizontal directions; confirm exact spacing with a structural engineer for the specific project.
3. Do not extend the 38 mm OD jack spindle past its 450 mm usable travel; place sole boards under jacks on soft or disturbed ground.
4. Inspect grating treads and landing panels regularly for corrosion, distortion, or weld cracking; withdraw and replace any damaged panel before further use.
5. 250 kg/m² is the maximum design UDL for both stair flights and landings; a structural engineer must confirm that actual site loading — personnel, tools, materials, and dynamic effects — stays within this figure.

Notes & Disclaimers

1. This datasheet is issued for information, reference, and quotation purposes. Project-specific structural calculations and certified drawings are prepared separately — contact Prime Steeltech for engineering support on a given project.
2. Dimensions are in mm unless stated otherwise. Weights and quantities are subject to ±3% tolerance.

Contact & Further Information

PRIME STEELTECH (I) PVT. LTD.

Office No. B-6, Tuljai CHS Ltd., Kasturba Cross
Road No. 4,
Borivali (East), Mumbai · 400 066, Maharashtra, India

Tel.: 86550 10046 | Mob.: +91 93245 60020
enquiry@primesteeltech.co.in
primesteeltech.com

Dharmesh Bhatt

Founder Director
Mob: +91-93223 21421 ·
dharmesh@primesteeltech.co.in

Jayraj Bhatt

Director · Business Development
Mob: +91-93931 61504 · jayraj@primesteeltech.co.in